X-ray Survey Catches an AGN Changing Its Look

Einstein Probe spotted a new X-ray source whose broad Balmer lines strengthened within four years.

arXiv:2610.30959 — Changing-look AGN EP J062132.0+623837 at z = 0.084

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arXiv:2610.30959 doi:10.48550/arXiv.2610.30959

Background

Changing-look AGNs shift between spectral types as their accretion and broad emission lines vary dramatically. Most known cases were found by optical surveys, not X-rays.

Wide-field time-domain X-ray surveys have barely been exploited to find changing-look AGNs systematically.

What this paper is selling

Einstein Probe's WXT discovered the source, and follow-up X-ray, optical, and mid-infrared data confirm a Type 1.8-to-1.5 transition. It shows X-ray surveys can complement optical searches for these objects.

Key results to remember

- X-ray luminosity of 3–9 × 10^43 erg/s with power-law slope Γ ≈ 1.5.
- Broad Balmer lines strengthened markedly between 2021 DESI and 2026 MMT spectra.
- Optical and mid-infrared light curves show dramatic brightening around 2010.

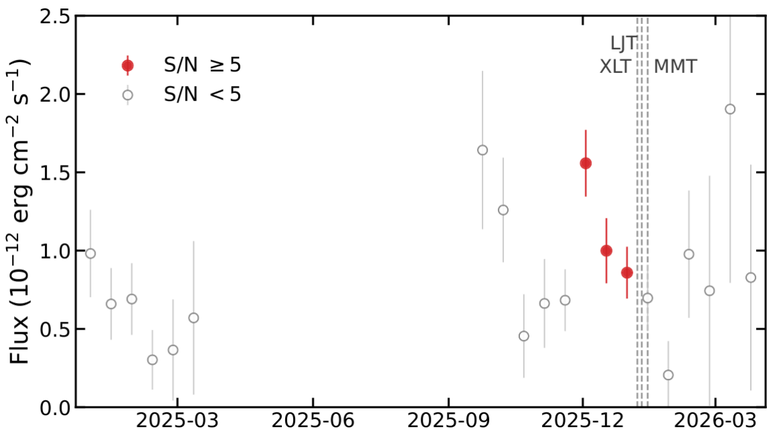


Figure 1 | Einstein Probe's 14-day binned X-ray light curve shows the source emerging in survey data, with spectroscopic follow-up epochs marked.

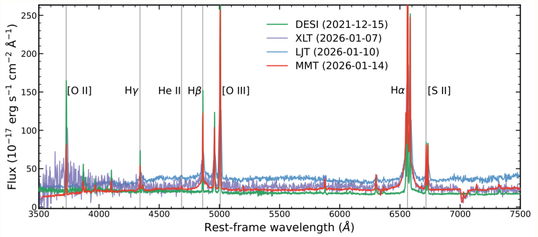


Figure 4 | Stacked rest-frame spectra reveal the broad Balmer lines growing stronger between the 2021 DESI and 2026 MMT observations.